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Magnetostratigraphy of the Kavur Tepe section (southwestern Turkey): A magnetic polarity time scale for the Norian

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ABSTRACT

We present the magnetostratigraphy of the late Triassic pelagic Kavur Tepe section from the Taurides (southwestern Turkey). The section investigated is about 30 m thick and consists of fine-grained whitish limestones. The biostratigraphy, based on the conodont zonation, gives an age from the Lacian 2 zone (lower Norian) to the Sevatian 2 zone (upper Norian). The paleomagnetic analyses indicate very favourable behaviour, with two magnetic components which can be isolated both by thermal and AF demagnetization. The magnetization is carried by magnetite and goethite, which are clearly evident in IRM experiments. The first component of normal magnetic polarity has roughly the direction of the present-day field. The high unblocking temperature component shows directions of both polarities. In total, fifteen magnetic polarity intervals are observed within the Kavur Tepe section. The characteristic calculated Fisherian mean direction is particularly tightly grouped: $D = 220.2°$ and $I = 32.1°$, with $K = 40.9$, $\alpha_{95} = 1.7°$ and $N_{samp} = 179$ (after bedding correction). Polarity correlations with two other sections in Turkey and in Austria (Northern Calcareous Alps) show that the Kavur Tepe sediments were deposited in the southern hemisphere. By combining the magnetostratigraphic results from the Bolücektasi Tepe [1] and the Kavur Tepe sections, an almost complete Norian magnetic polarity sequence is established, showing 30 magnetic intervals that are well calibrated with the conodont and ammonoid zonations. A magnetic reversal frequency close to about 2.7 reversals/Myr is derived for the Norian. This medium frequency is in agreement with the hypothesis of a long time constant (150–200 Myr) in the changes of the magnetic reversal frequency since the Kiaman reversed superchron.

1. Introduction

In a recent paper, we [1] presented the magnetostratigraphy of the late Triassic Bolücektasi Tepe section in the western Taurides of Turkey. In this section of light pink and whitish pelagic limestones a sequence of 39 magnetic polarity intervals has been obtained, encompassing the lower Carnian, the uppermost Carnian and parts of the lower and middle Norian. From this magnetic sequence, which is well calibrated by the conodont zonation, we proposed that both the Carnian and the Norian were periods of medium magnetic reversal frequency, with close to 3 reversals/Myr. This result seems comparable to the reversal frequency estimate derived from the magnetostratigraphy of Norian–Rhaetian marine sediments from ODP Leg 122 [2]. However, it notably contradicts the low-frequency estimate that has been derived from the scarce available late Triassic magnetostratigraphic results obtained in continental sedimentation environments [3–8]. In particular, the late Triassic does not appear as the period of predominantly normal magnetic polarity that was previously suggested [9]. Uncertainties about the completeness of sedimentary records in continental environments, which often have a poorly defined biostratigra-

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phy, may easily explain this discrepancy. By considering the late Triassic magnetic reversal frequencies derived from the Bolücektasi Tepe section, together with the available and reliable magnetostratigraphic results for the middle and lower Jurassic [10,11] and those for the early Triassic

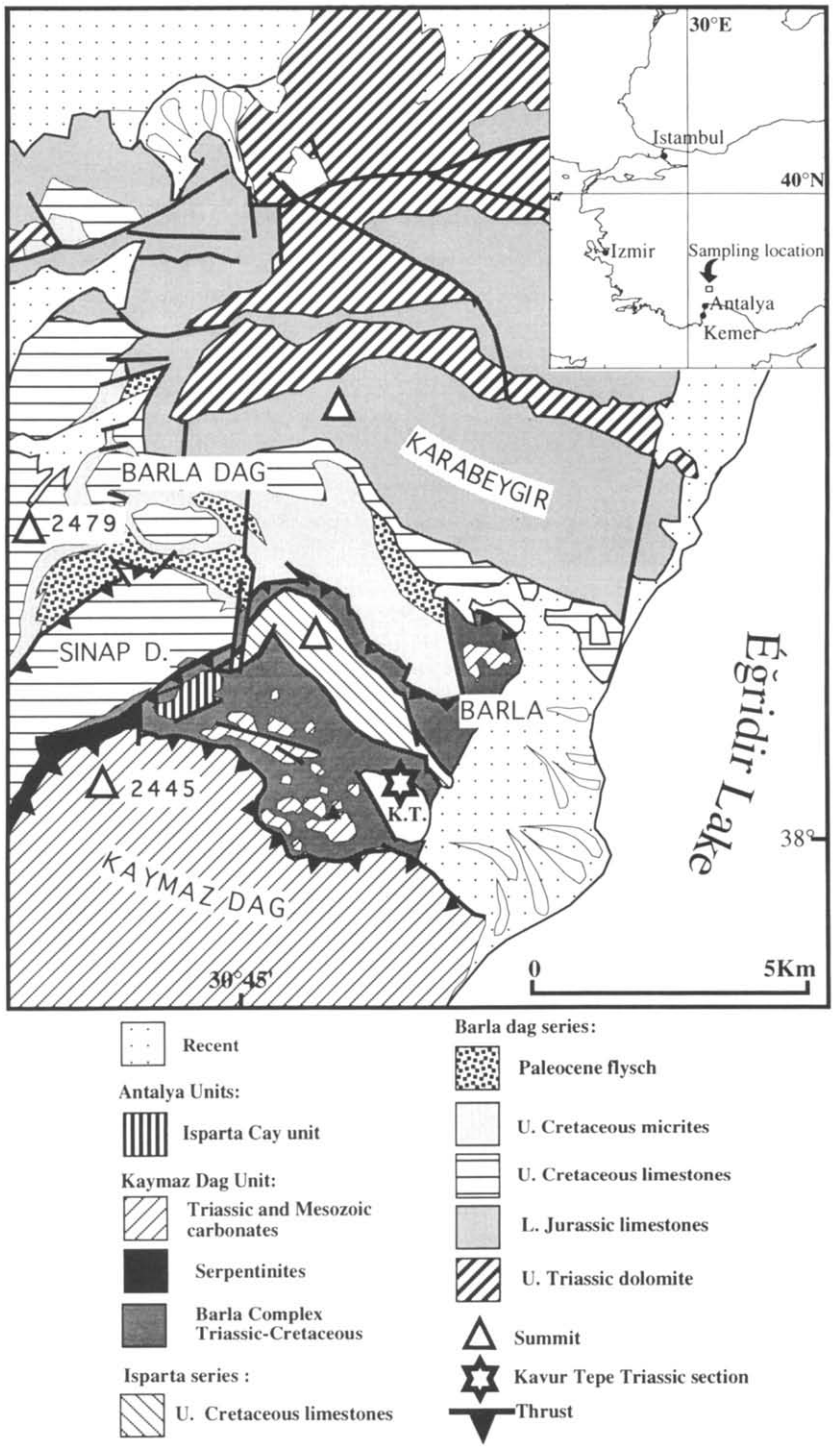


Fig. 1. Location map of the Kavur Tepe section and simplified geological and structural map of the sampling area [after 22].

and late Permian [12–14], we have tested the hypothesis of a long time constant of about 150–200 Myr in the changes of the reversal frequency since approximately 320 Myr [15–17]. Although the magnetic sequence deduced from the magnetostratigraphic results is still very discontinuous, a rough agreement has been observed between the model and the data [1]. However, this synthesis also illuminated both the problems due to the magnetostratigraphic data themselves (paleomagnetic reliability, biostratigraphic control, and amount of data) and those due to the uncertainties regarding the radiometric ages of the early Mesozoic geological stage limits (and hence uncertainties regarding the duration of these stages). A good example is the early Triassic, for which the duration changes by a factor of two depending on the published time scale [e.g., 18–20; see also 1]. Our first study thus demonstrated the importance of obtaining new magnetostratigraphic data to better ascertain the magnetic reversal frequency change model for these periods.

In this paper, we present the magnetostratigraphy of the pelagic Kavur Tepe section in southwestern Turkey. The age of this section, lower to upper Norian (late Triassic), provides the possibility of completing the magnetic polarity sequence during the late Triassic and of better constraining the magnetic reversal frequency during this period.

2. Geological setting and description of the section

The Kavur Tepe is a small summit (1318 m) located in the western Taurides of southwestern Turkey (38°0′, 30°5′) on the western shore Lake Egridir, about 1 km south of the village of Barla (Fig. 1). It is a small oval rounded carbonate klippe with its long axis parallel to the strike. The beds dip to the southwest, gently on the northern flank (approximately 30°) and steeply on the southern side (approximatively 60°). The lithologies consist of Triassic pelagic and Jurassic–Cretaceous neritic limestones [21] tectonically overlying the so-called Barla complex [22]. The latter, which is upper Cretaceous in age, is of varying lithology, but consists dominantly of sandstones. As suggested by Gutnic et al. [22], the Barla complex could correspond to 'wildflysch' detached from its basement, which has, however, never been observed.

The Kavur Tepe klippe and other nearby carbonate klippes belong to the Kaymaz Dag unit [22]. Although our knowledge of the stratigraphy of this unit is incomplete, three formations may be distinguished: (1) the Paleozoic (late Permian) carbonates, (2) the lower–middle(?) Triassic Bedre formation, and (3) the Kaymaz Dag carbonates. The Kaymaz Dag carbonates have been lithostratigraphically related to the Upper Antalya Nappes, which have been described in more detail in the Kemer area close to Antalya, on the southwestern coast of Turkey [22–24].

The carbonate sequence of Kavur Tepe begins with reddish to brownish pelagic limestones containing a Carnian ammonite bed [22]. The Norian sequence consists of two different carbonate rock packages. The investigated section of fine-grained, thin- to medium-bedded whitish limestones of the Hallstatt type is approximately 30 m thick. This sequence contains scarce ammonites and several age-diagnostic bivalve layers. The biostratigraphy, which has been established by one of us (LK), is based on conodonts, which provide a rich and continuous faunal record throughout the section. It appears that about 40 collected samples define seven zones, based on the same taxonomic concept as in the Bolücektasi Tepe section [1], from the Lacian 2 zone (lower Norian) to the lower part of the Sevatian 2 zone (upper Norian).

The Hallstatt limestones are overlain by 150 m of whitish, medium to thick bedded, fine- to coarse-grained limestones that contain slope and shallow-water debris (crinoids, lithoclasts and ooids). Conodonts are rare in this sequence and only a Sevatian 2 (and Rhaetian?) age can be tentatively suggested. Within the uppermost 15 m, an increasing input of lithoclasts and reworked frame-building organisms indicates the nearby location of a prograding shallow-water platform, as in other parts of the Upper Antalya Nappes. This sequence is lithostratigraphically similar, but slightly younger, than the Allodapic member of the Bolücektasi Tepe section [1]. Considering the poorly defined biotratigraphic control, a magnetostratigraphic study has not been performed in this part of the section. Finally, approximately 50 m of carbonate breccias with upper(?) Jurassic

and Cretaceous shallow-water pebbles unconformably cover the late Triassic sequence. Interlayers consisting of thin-bedded packstones and grainstones have been dated by foraminifera as Cretaceous in age (middle–upper Cenomanian; I. Bodrogi, pers. commun.). The breccias are topped by 2 m of greenish-grey conglomerate with various limestone and ophiolite clasts. This conglomerate most probably represents the well-known Maastrichtian ophiolitic olitostrome of the Antalya Nappes [24].

In our previous study [1], we compared three Triassic time scales [18–20]. Considering the paucity of the available radiometric ages for the Triassic, we proposed an alternative procedure to estimate the ages of the Triassic stage boundaries which consisted in using the rough concept of equal duration for ammonoid zones between three tie points (Jurassic–Triassic boundary at 205 Ma, Aniso-Ladinian boundary at 235 Ma and Permo-Triassic boundary at 250 Ma). The results are given in fig. 3 in Gallet et al. [1]. Following this concept, the approximate mean duration per ammonoid biochron during the Norian would be 1.5 Myr, giving a total duration of about 8–9 Myr for the studied section. During two fieldtrips, in 1990 and 1991, about 220 paleomagnetic cores were collected, providing a mean time resolution of about 40 kyr/sample.

3. Paleomagnetic analyses

Most of the paleomagnetic analyses were performed with the three-axis CTF cryogenic magnetometer in the shielded paleomagnetic laboratory of the Scripps Institution of Oceanography (La Jolla, USA). In addition, some analyses were also carried out, using the same brand of magnetometer, at the *Institut de Physique du Globe de Paris*. Although some of the samples were demagnetized by alternating fields (AF) or by mixed demagnetizations (thermal up to 150°C, and then

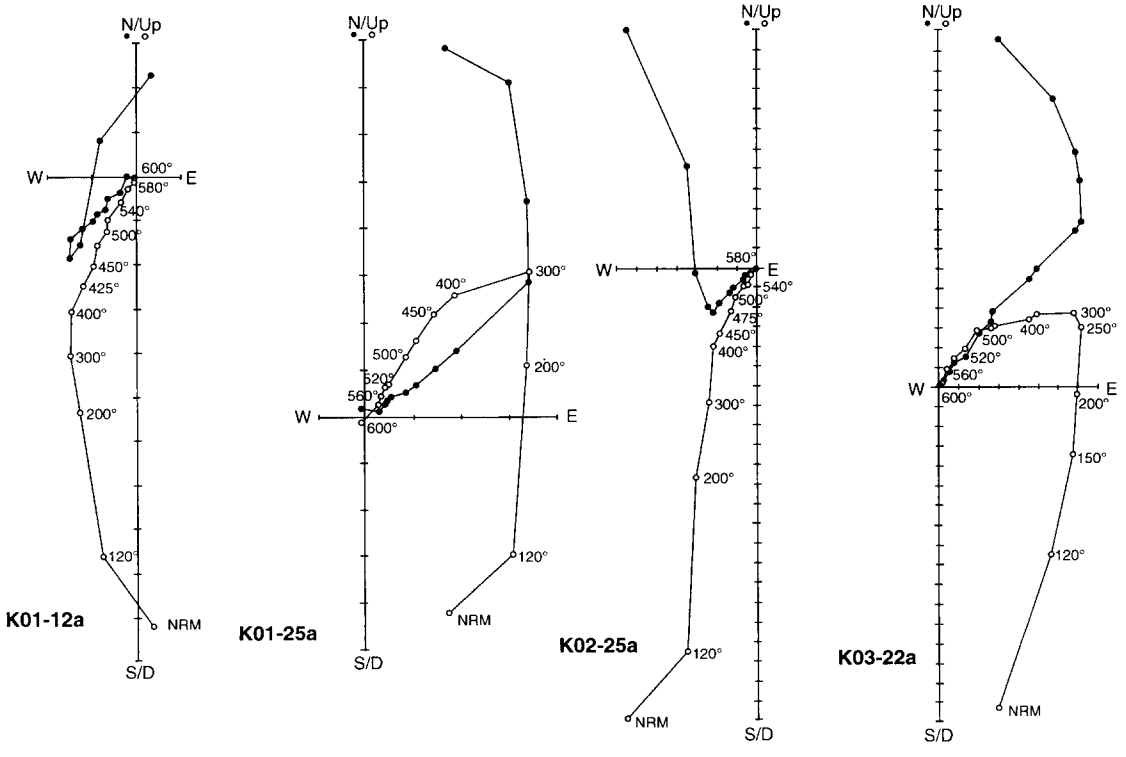


Fig. 2. Vector plots of progressive thermal demagnetizations of samples well distributed through the section. The closed symbols refer to the horizontal plane, the open symbols to the vertical plane. All diagrams are in stratigraphic coordinates, with scale bars 10^{-5} A/m apart. These diagrams clearly show two well-defined magnetic components with an overlap in the medium-temperature range. The characteristic component is of normal or reversed magnetic polarity depending on the samples.

AF), thermal demagnetizations were most often chosen to isolate the different magnetic components. In this case, approximately fifteen heating steps were used, with measurements of the bulk susceptibility after each step to check the stability of the magnetic mineralogy during the heating process. The magnetic components were selected using a principal component analysis (PCA) routine [25].

The intensity of the natural remanent magnetization (NRM) measured before any demagnetization is homogeneous throughout the section, with an arithmetic mean of 0.17 mA/m $\pm$ 5.6%. The bulk susceptibility is very weak and has mostly negative values ($> -10^{-5}$ SI). The stability of these values within the section probably reflects the very homogeneous whitish lithology.

Examples of characteristic thermal demagnetizations of the NRM plotted on orthogonal vector diagrams [26] are shown in Fig. 2. These plots indicate a very favourable paleomagnetic behaviour throughout the Kavur Tepe section. Two magnetic components are easily isolated. The first one, in the low- to medium-temperature range (from 120° to 250°–300°C), is always of normal magnetic polarity. The second component is isolated in the higher temperatures, from approximately 500° to 600°C, and has either a normal or a reversed magnetic polarity. An overlap between these two components clearly exists between 300° and 400°C. Measurements of the bulk susceptibility after each thermal step show that the values remain very stable during the heating, and thus that no change in the magnetic mineralogy occurs during the treatment (Fig. 3).

When we compare demagnetization diagrams from thermal and AF treatment (Fig. 4), we observe essentially the same behaviour, with two well-defined components and an overlap in the medium-temperature and coercivity range (about 20–30 mT). A small component of normal polarity has still not been removed by the end of AF demagnetization (achieved in fields of 100 mT) (Fig. 4). This component, which clearly has the direction of the component removed both in low alternating fields and in low-temperature heating steps, is probably carried by goethite.

To better characterize the magnetic carriers, we have thermally demagnetized a three-axis differential IRM (0.12, 0.4 and 2.3 T) following Lowrie's method [27]. Very homogeneous behaviour is obtained (Fig. 5). The magnetization is mostly carried by a low-coercivity mineral with an unblocking temperature of approximately 560°C.

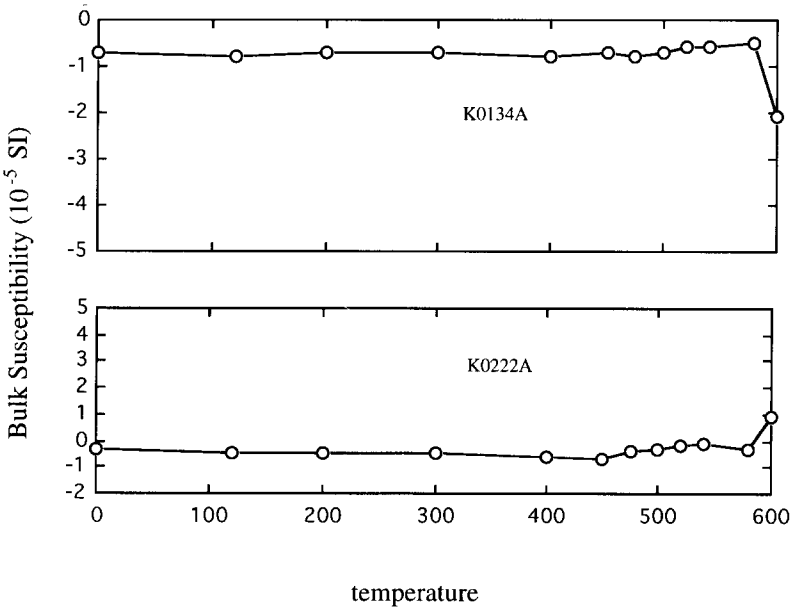


Fig. 3. Evolution of bulk susceptibility during the thermal demagnetizations for two samples. The values remain roughly constant untill the end of the demagnetization, showing no (or little) change in the magnetic mineralogy during the thermal treatment.

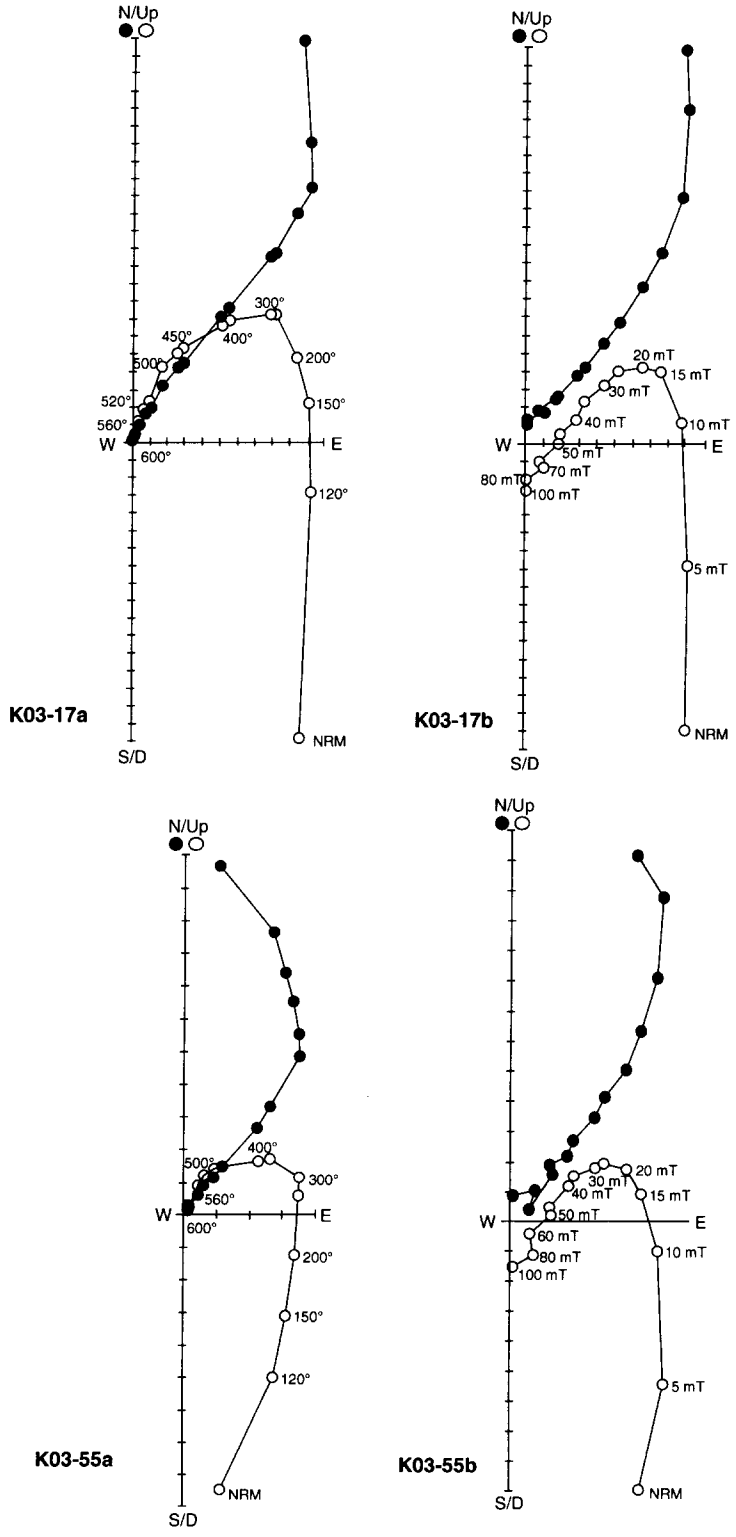


Fig. 4. Comparison of orthogonal projections of AF and thermal demagnetizations for two cores. Same conventions as in Fig. 2, and scale bars 10^{-5} A/m apart. Identical behaviour is observed during the two treatments. However, a small component, which has in both cases the direction of the low-stability component, cannot be removed by AF demagnetization.

A mineral from the (Ti) magnetite family is thus probably the magnetic carrier. The medium (i.e., 0.4 T) coercivity decay curve is negligible. However, the hard coercivity curve describes a very characteristic behaviour, with a significant drop between NRM and 120°C. Goethite, which we suspected from the AF demagnetization, is then evident from the IRM experiment. Finally, no hematite or iron sulphides are evident in the samples.

4. Paleomagnetic directions and magnetic stratigraphy

The directions of the low unblocking temperature component are plotted on an equal-area projection in Fig. 6a. These directions are well grouped and are roughly parallel to the present-day field direction. The Fisherian [28] mean direction (in geographic coordinates) is $D = 358.8°$, $I = 55.1°$, with $K = 33.4$ and $\alpha_{95} = 2.7°$ (established from 82 samples; component 1, Table 1). For comparison, the present-day field direction is $D = 3.5°$, $I = 54.9°$, and the geocentric axial dipole direction gives $I = 57.4°$ ($D = 0°$).

The characteristic high unblocking temperature components (ChRM) are well grouped in the two polarity states (Fig. 6b). The directions of both magnetic polarity have a good circular (Fisherian) distribution. However, the calculated Fisherian mean directions are not exactly antipodal with respect to inclination (component 2, Table 1). The reason for this difference is unclear. We could suggest, with caution, that the overlap between the two components might bias the final direction. The characteristic Fisherian mean direction which is determined despite the distribution is particularly tightly grouped: $D = 220.2°$, $I = 32.1°$, with $K = 40.9$, $\alpha_{95} = 1.7°$ and $N_{samp} = 179$ (after bedding correction, Table 1). No signif-

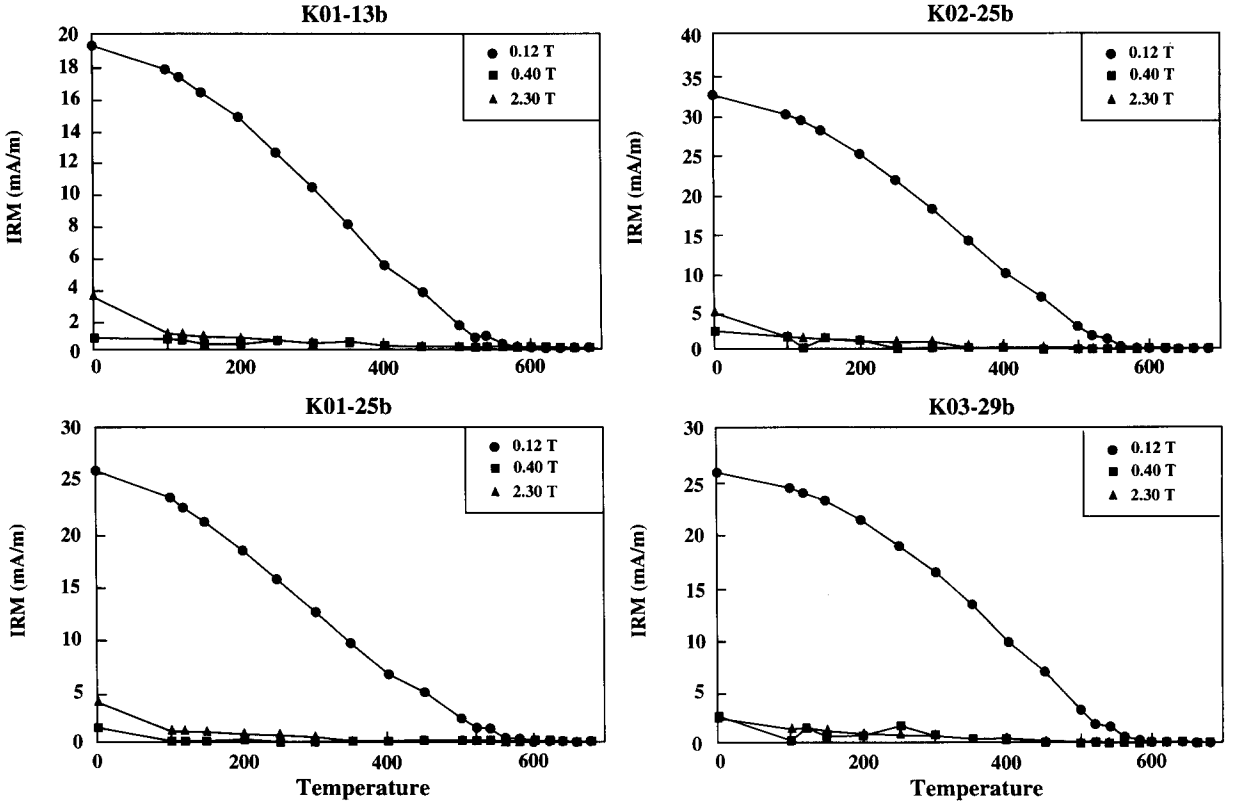


Fig. 5. Thermal demagnetizations of a three-axis differential IRM for four representative samples through the section. The results indicate that the magnetization is mostly carried by a mineral, probably in the magnetite family, of low coercivity, with an unblocking temperature of 560°–580°C. Moreover, a mineral of high coercivity is demagnetized at 120°C. This is probably goethite as is also evident from the AF demagnetization (see Fig. 4).

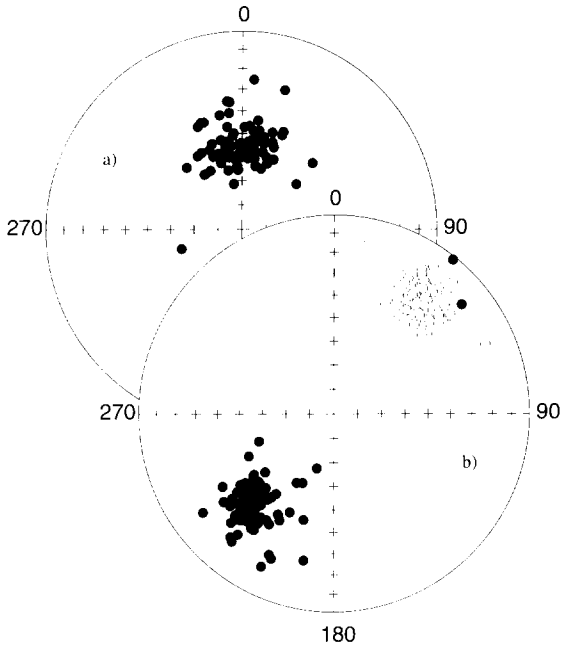


Fig. 6. Equal-area stereographic projections of the magnetic components isolated throughout the Kavur Tepe section. Closed symbols refer to directions in the lower hemisphere, open symbols to directions in the upper hemisphere. (a) Magnetic components of low unblocking temperature (i.e., < 250°–300°C). The directions closely parallel the present-day field direction; few samples seem to be clearly misoriented. (b) Characteristic components of high unblocking temperature (i.e., obtained between 500°C and 600°C). These directions are tightly grouped in the two polarity states.

icant trend in the magnetic directions is observed within the Kavur Tepe section (Fig. 7).

From the ChRM directions, a well-defined magnetostratigraphic signature with fifteen magnetic intervals is obtained within the Kavur Tepe section (Fig. 7). The latitude of the virtual geomagnetic pole (VGP) has also been computed for each direction and the deduced magnetic polarity zones are plotted on the right-hand side of the figure, considering a southern hemisphere origin for the section. Following Alvarez et al. [29], the polarity zones are designated by letters in ascending alphabetic order, with the prefix KT (Kavur Tepe). The normal and reversed polarities are indicated respectively with plus and minus signs. Some zones (C +, F − and H −) are made up of several stratigraphically thinner magnetic subzones (additionally labelled with a number). Despite the relatively high sampling density, two intervals in the upper Norian are defined by only one sample (short horizontal lines in Fig. 7). However, the paleomagnetic directions obtained from these samples are clear. Finally, two samples have probably recorded an intermediate direction during a magnetic reversal.

5. Discussion

This study has been conducted with the aim of completing the Norian magnetostratigraphic sequence already obtained from the Bolücektasi Tepe section. In this latter section, the upper lower Norian (Lacian 3 zone) and the upper Norian (Sevatian 1 and 2 zones) are missing. In Kavur Tepe, these zones are present, although the upper part of the Sevatian 2 zone is unsuitable for study due to faulting. Integration of these new results to establish a composite 'reference' Norian magnetic polarity sequence depends on a successful magnetostratigraphic correlation between the two Turkish sequences.

TABLE 1

Mean paleomagnetic directions and VGP of the Kavur Tepe section

	N_{samp}	Geographic coordinates	Stratigraphic coordinates
Component 1	82	$D = 358.8$, $I = 55.1$, $K = 33.4$, $\alpha_{95} = 2.7$	
Component 2			
1st polarity	91	$D = 219.4$, $I = 59.0$, $K = 35.4$, $\alpha_{95} = 2.5$	$D = 221.5$, $I = 37.4$, $K = 50.2$, $\alpha_{95} = 2.1$
2nd polarity	88	$D = 32.8$, $I = -48.5$, $K = 31.1$, $\alpha_{95} = 2.7$	$D = 39.1$, $I = -26.7$, $K = 49.4$, $\alpha_{95} = 2.2$
General	179	$D = 215.8$, $I = 53.9$, $K = 28.9$, $\alpha_{95} = 2.0$	$D = 220.2$, $I = 32.1$, $K = 40.9$, $\alpha_{95} = 1.7$
VGP			Lat. = −23.0, Long. = 72.5, $d_p = 1.1$, $d_m = 1.9$, paleolatitude = ±17.4

The mean direction vector for Kavur Tepe, which clearly shows a negative inclination when its direction points toward the geographic North, may indicate a southern hemisphere latitude of deposition. Negative inclinations (magnetic vector pointing to the north) would thus correspond to normal polarities. However, given the complex geological setting of the Antalya Calcareous Nappes, a large-scale rotation may also possibly have occurred during the nappe emplacement, giving an opposite interpretation in terms of polarities. One way to solve this problem is to compare the magnetostratigraphy from Kavur Tepe with that obtained in Bolücektasi Tepe, as parts of the two sections cover the same time interval. Indeed, most of the middle Norian (Alaunian zone) is common to both sections. Although Bolücektasi Tepe also belongs to the Antalya Nappe System, its northern hemisphere latitude of formation is ascertained by an evolution in paleolatitude that is coeval with that of Gondwana during the late middle/late Triassic [in prep.]. Moreover, we have recently conducted a magnetostratigraphic study of a lower Carnian (Julian 1 zone) section from the Northern Calcareous Alps (Mayerling, Austria). This region was clearly situated in the northern hemisphere during the late Triassic [30–32]. Anticipating the related results [in prep.], a satisfactory correlation in terms of magnetic polarities is found between the Bolücektasi and Mayerling sections, which also confirms the northern position of Bolücektasi Tepe. The comparison between the two magnetic sequences of Turkey is shown in Fig. 8. From bottom to top, the middle part of the lower Norian shows a normal polarity, the base of the Alaunian is reversed, a long normal polarity interval can be observed from at least the upper part of the Alaunian 1 zone to the lower part of the Alaunian 3 zone, and the top of the common part shows a reversed polarity. From this figure it seems clear that a satisfactory correlation is obtained for the common part of the two sections. This result thus suggests that Kavur Tepe sediments were probably deposited in the southern hemisphere (at a paleolatitude of 17.4°S, Table

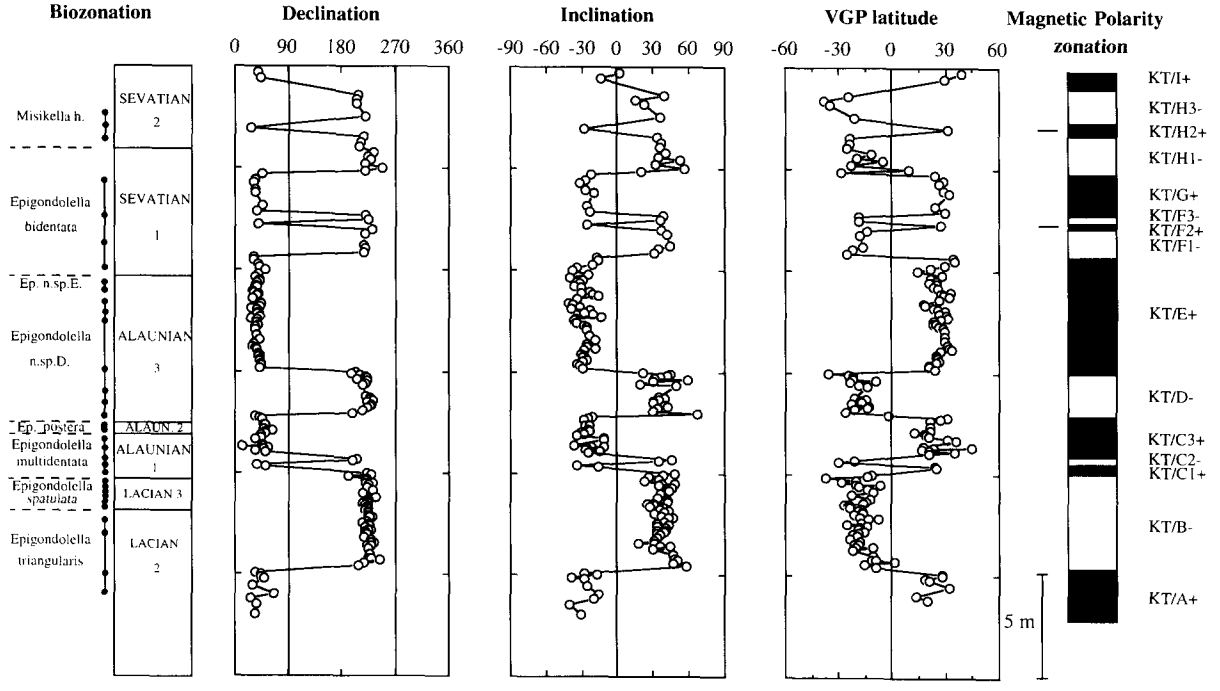


Fig. 7. Biostratigraphy and magnetic stratigraphy of the Kavur Tepe section. The biostratigraphy is established from conodont zonations. No trend in the magnetic directions is observed. A sequence of fifteen magnetic intervals is deduced and shown considering a southern hemisphere origin for the section. Following Alvarez et al. [29], the polarity zones are designated by letters in ascending alphabetic order. Two of these intervals, which are probably of short duration, are defined by only one sample (short horizontal lines).

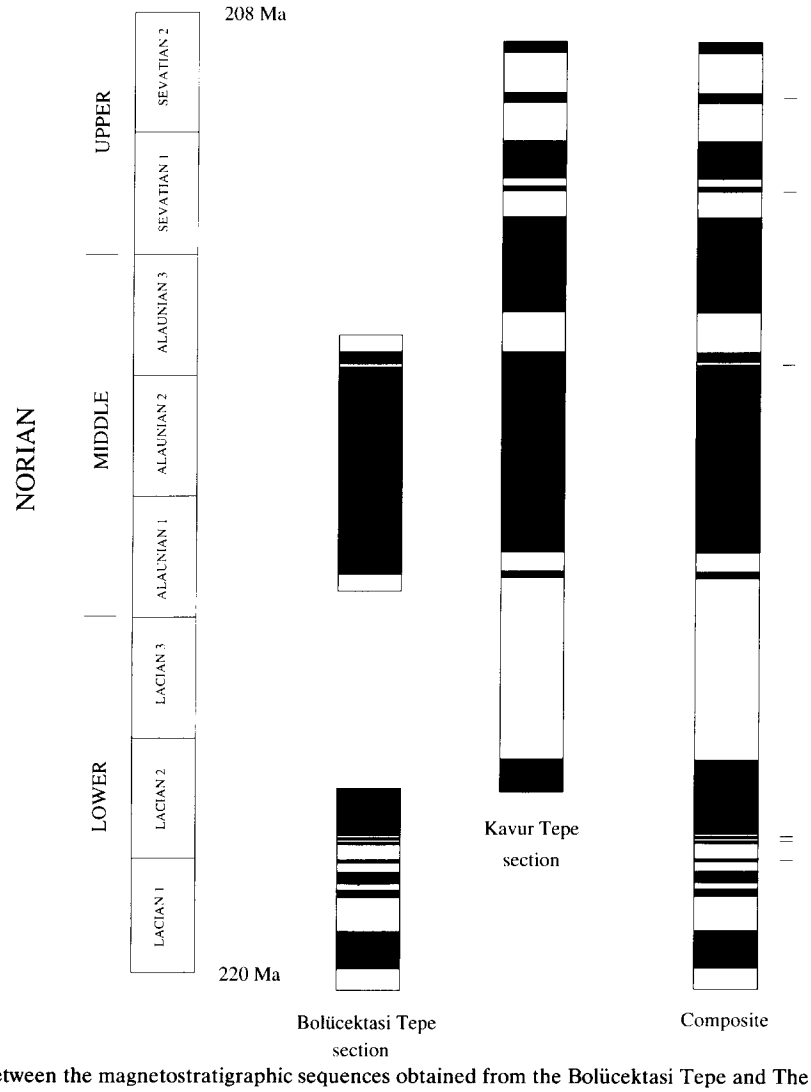


Fig. 8. Comparison between the magnetostratigraphic sequences obtained from the Bolücektasi Tepe and The Kavur Tepe sections. The numerical time frame has been estimated using the concept of equal duration for the ammonoid zones [1]. The polarity sequence of the Kavur Tepe section is given according to a southern hemisphere origin for this section. The good correlation between the two sections for the three Alaunian zones (middle Norian) indicates their origins from opposite hemispheres and by deduction, the southern hemisphere origin of the Kavur Tepe section (see text). A composite Norian geomagnetic polarity sequence is shown on the right of the figure. The short horizontal lines on the right indicate the intervals defined by only one sample.

1). A sketch map of the Tethys area is shown in Fig. 9. Laurasia and Gondwana have been replaced to their ancient position using the compilation of Van der Voo [33] for Eurasia and a compilation of South American and African poles made by Théveniaut et al. [in prep.] for Gondwana. The mean Gondwanian pole for this period does not differ significantly from previous compilations [34,35]. While the paleoposition found for Bolücektasi Tepe indicates that this succession was formed near the northern edge of the Arabian promontory, the paleolatitude found for Kavur Tepe shows a position of formation close to the northern margin of India during the late Triassic. Both successions are sealed by detrital ophiolitic sediments of upper Cretaceous age [24], indicating that, at the latest, they joined together at this time. A differential latitudinal motion of

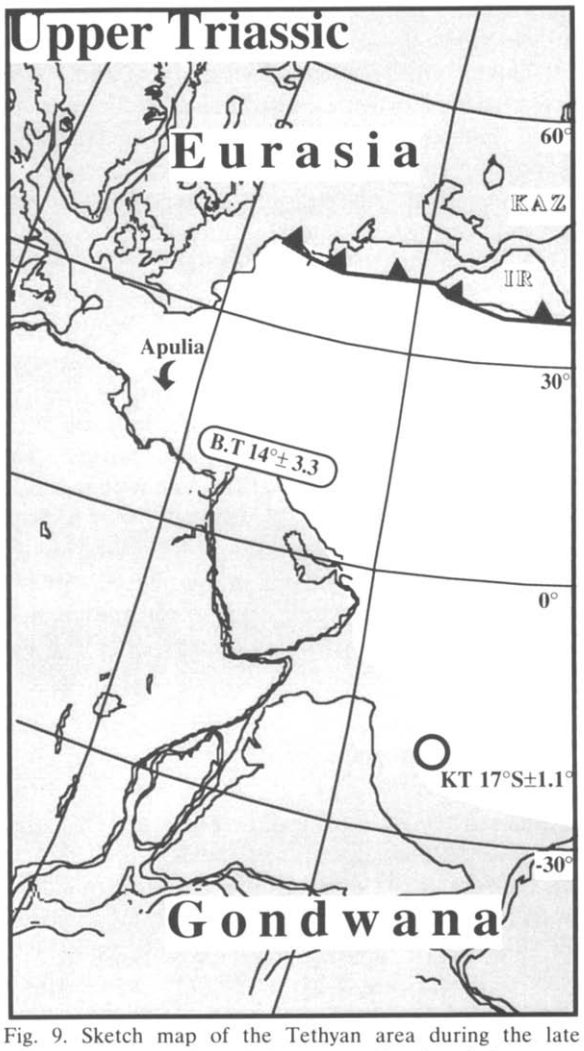


Fig. 9. Sketch map of the Tethyan area during the late Triassic with Laurasia and Gondwana placed at their respective paleopositions. The paleolatitudes (and their confidence limits) for the two Turkish sections of the Antalya Nappes are also indicated: *BT* is the Bolücektasi Tepe section and *KT* is the Kavur Tepe section. Note that the magnetostratigraphic correlation discussed in the text implies a significantly different latitude of formation for these two units.

some 2500 km is found, implying a mean convergence rate of about 2.5 cm/yr between the Triassic, when the fragmentation of parts of the northern Gondwanian margin occurred, and the Cretaceous. Although the tectonic setting of the Tethyan area is not directly comparable to that of western America, this situation is reminiscent of that observed in the Franciscan Complex limestones of northern California [36–38], where blocks of different origin are now located near each other.

By combining the two sequences, three—and perhaps five—magnetic intervals are obtained within the Alaunian 3 zone. The interval defined by only one sample in the Bolücektasi Tepe section has not been found in Kavur Tepe. A similar problem is also observed in the Alaunian 1 zone, where one interval defined by two samples in Kavur Tepe is missing in Bolücektasi Tepe. In both cases, these intervals are probably of short duration and thus it will be difficult to observe them due to the sampling process. New results are obtained for the Lacian 3 zone, where only a reversed magnetic polarity is observed in Kavur Tepe. This reversed magnetic interval, which is also present in the upper part of the Lacian 2 zone, is missing in the Bolücektasi Tepe section due to faulting. The Kavur Tepe results for the Sevatian zones (the Sevatian 2 zone is not complete) show nine polarity intervals, but two of them, which are probably of short duration, are poorly defined by only one sample.

In order to summarize the magnetostratigraphic results obtained from Bolücektasi Tepe and Kavur Tepe, we have established a composite Norian magnetic sequence and used a linear time scale estimated from the hypothesis of an equal duration for the ammonoid zones [1]. In total, 30 intervals are observed from the Lacian 1 zone to the lower part of the Sevatian 2 zone. Six intervals are defined by only one sample and should be confirmed by additional data. If all the intervals are considered, a mean magnetic reversal frequency of about 2.7 reversals/Myr is deduced. If we exclude the one-sample intervals, this reversal frequency decreases to 2.2 reversals/Myr. These estimates are similar to those that were deduced from the Bolücektasi Tepe section only. The discussion we presented in this study [1] concerning the fluctuations in magnetic reversal frequency since the end of the Kiaman reversed superchron thus remains unaffected. The hypothesis of a long time constant in the changes of the magnetic reversal frequencies [15,16] seems to be roughly confirmed by the available magnetostratigraphic data.

Recently, Witte et al. [8] have suggested a composite geomagnetic polarity reference late

Triassic sequence from the paleomagnetic results obtained in the Newark Basin. They propose a sequence of seven magnetic intervals during the Norian (including the Rhaetian). No correlation is seen with the sequence we show in this study. As previously pointed out, problems associated with stratigraphic completeness and biostratigraphic zonation are common in continental sedimentation environments. These difficulties may therefore explain the discrepancy between the two magnetic polarity sequences. Similar problems are also suggested for the magnetostratigraphic sequence obtained from the Redonda Formation which constitutes the upper part of the Chinle Formation in eastern New Mexico [3]. This formation is dated at middle–upper Norian [e.g., 39]. The relatively long reversed magnetic interval which is obtained cannot be confidently correlated with one interval from our sequence. Moreover, in addition to the stratigraphic problems we also note a large difference in sampling density between the sequences derived from the Newark Basin and the Turkish results. The Norian sequence proposed by Witte et al. [8] was established from approximately 22 paleomagnetic sites (or stratigraphic horizons). This means that the mean time resolution given by this sampling is of the order of 0.5 Myr/site. By comparison, the sequence derived from the Turkish results was established from 315 individual samples, giving a mean time resolution of approximately 0.04 My/sample. Such a large difference in resolution probably suggests that the intervals found in the Newark Basin are the concatenation of several shorter magnetic intervals. Similar effects have been illustrated by simple statistical experiments [40,41]. This example of discrepancy also clearly illustrates the problems encountered in magnetostratigraphy, particularly when a reference geomagnetic time scale is tentatively established.

6. Conclusion

The mean direction obtained in Kavur Tepe ($D = 220.2°$, $I = 32.1°$, with $K = 40.9$ and $\alpha_{95} = 1.7°$) together with the magnetostatigraphic correlation between the two Turkish sections shows that Kavur Tepe was probably formed in the southern hemisphere. Units of very different origin are now found in the Antalya Nappes System as a result of the successive collision of migrating blocks through the Tethys area.

The magnetostratigraphic study of the Kavur Tepe section provides a sequence of fifteen magnetic intervals from the lower Norian (Lacian 2 zone) to the upper Norian (Sevatian 2 zone). A clear magnetostratigraphic correlation is observed with the middle Norian polarity sequence already obtained from the Bolücektasi Tepe section [1]. Combining the two sequences, a composite geomagnetic polarity sequence, with 30 intervals well calibrated in time with the conodont zonation, is proposed for most of the Norian. Six of these intervals, which are probably of short duration, are defined by only one sample. The moderate magnetic reversal frequency that is estimated for the Norian (2.7 reversals/Myr with all the intervals and 2.2 reversals/Myr excluding the poorly defined intervals) is in agreement with the hypothesis of a long time constant in the changes of the magnetic reversal frequency since the end of the Kiaman.

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